

Relative Amplitude TD Exploration: Reward-Scale Robust Adaptive Exploration and Approximator-Dependent Performance in Value-Based Reinforcement Learning

Ihzaz Abrar Sadik^a

^a*Department of Computer Science and Engineering, BRAC University, Dhaka, Bangladesh*

Abstract

Keywords: reinforcement learning, exploration, temporal-difference error, function approximation, VDBE, Double DQN, reward scaling, robust evaluation

1. Introduction

2. Background and Related Work

2.1. Value-based control and TD error

2.2. Exploration schedules and adaptive exploration

2.3. Evaluation under seed variability

3. Relative Amplitude TD Exploration

3.1. Running amplitudes

3.2. Positive reward-scale property

3.3. Computational cost and neural integration

4. Experimental Methodology

4.1. Environments and interaction budgets

Four environments with continuous state spaces, discrete actions but varying structure have been used. They all give the exploration rules different kinds of feedback. This helps in assessing the exploration methods across a diverse range of scenarios. The ‘Gymnasium’ package in Python, contains

these environments and were used by parsing them from it. Summary of all the environments are listed in table 1.

Table 1: Summary of the benchmark environments.

Environment	Obs. Dim.	State Dime.	$ A $	Final Budget
MountainCar-v0	2	2	3	150,000
CartPole-v1	4	4	2	100,000
Acrobot-v1	6	4	3	100,000
LunarLander-v3	8	8	4	300,000

Environment Details

1. MountainCar-v0

- **Brief Intro:** An underpowered car in a valley between two hills, must reach a flag on the right hill. The engine is weaker than gravity.
- **Reward Structure:** -1 per step until goal
- **Significance:** Success is sparse, exploration needs to discover momentum in the opposite direction to the goal

2. CartPole-v1

- **Brief Intro:** A pole is hinged on a cart; it can be pushed right or left to keep the pole upright.
- **Reward Structure:** +1 for each surviving step
- **Significance:** Dense reward from the very first step

3. Acrobot-v1

- **Brief Intro:** A two-link pendulum hanging from a pivot, actuated only at the middle joint, must swing its tip above a given height.
- **Reward Structure:** -1 per step until goal
- **Significance:** Structured long-horizon control

4. LunarLander-v3

- **Brief Intro:** A lander must set down softly on a pad between two flags.
- **Reward Structure:** shaped+terminal
- **Significance:** Stochastic, high-dimensional, complex reward structure

4.2. *Linear approximator*

4.3. *Double DQN*

4.4. *Exploration baselines and frozen tuning*

4.5. *Greedy evaluation*

4.6. *Robust statistics*

5. Diagnostic Experiments

5.1. *Experiment A: Is raw TD-error magnitude a progress signal?*

5.2. *Experiment B: VDBE's TD-independent large- σ limit*

5.3. *VDBE fidelity ablation*

5.4. *Experiment C: positive reward rescaling*

6. Final 30-Seed Benchmark

6.1. *Linear Sarsa(λ)*

6.2. *Double DQN*

6.3. *Approximator-dependent ranking*

7. Discussion

7.1. *Scale robustness is not performance optimality*

7.2. *Why might the approximator matter?*

7.3. *What the VDBE diagnostics add*

7.4. *MountainCar under Double DQN*

8. Limitations

9. Conclusion

Data and Reproducibility Statement

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